

Petri nets: Liveness is monotonic in neutral cycles

Heorhii Lopattn

June 7, 2025

Solution

Assume that the marking M is live, and $M \leq M'$. We will prove the following statement that directly implies that M' is live:

For any σ , there exist σ_1, σ_2 s.t. $M' \xrightarrow{\sigma} M'' \xrightarrow{\sigma_1} M'_1 \geq M_1 \xleftarrow{\sigma_2} M$.

We will use induction on σ .

The base case is obvious, as $M \leq M'$. In the induction step we let $\sigma = \tilde{\sigma} t_i$ where t_i is a transition from p_i to $p_{i+1 \bmod n}$. From the inductive assumption we have $\tilde{\sigma}_1, \tilde{\sigma}_2$, s.t. $M' [\tilde{\sigma}] [\tilde{\sigma}_1] \geq M [\tilde{\sigma}_2]$

There are two different cases:

If t_i appears in $\tilde{\sigma}_1$, it is interchangeable with the rest of the transition, so we can rewrite $\tilde{\sigma}_1 = t_i \tilde{\sigma}_1'$. This means we can take $\sigma_1 = \tilde{\sigma}_1'$ and $\sigma_2 = \tilde{\sigma}_2$.

If t_i doesn't appear in $\tilde{\sigma}_1$, observe that $M' [\tilde{\sigma}_1] [t_i] = M' [t_i] [\tilde{\sigma}_1]$.

Since by inductive hypothesis, $M' [\tilde{\sigma}] [\tilde{\sigma}_1] \tilde{M}_1 \geq M [\tilde{\sigma}_2] \tilde{M}_1'$, we use liveness of $\tilde{M}_1'$ to obtain a firing sequence $\alpha = \alpha_1 t_i \alpha_2$. Obviously it also applies to $\tilde{M}_1$, as it is greater. It holds that

$$\begin{aligned} \tilde{M}_1 [\alpha] &\geq \tilde{M}_1' [\alpha] \\ \iff M' [\tilde{\sigma}] [\tilde{\sigma}_1] [\alpha] &\geq M [\tilde{\sigma}_2] [\alpha] \\ \iff M' [\tilde{\sigma} t_i] [\tilde{\sigma}_1] [\alpha_1 \alpha_2] &\geq M [\tilde{\sigma}_2] [\alpha] \end{aligned}$$

And we take $\sigma_1 = \tilde{\sigma}_1 \alpha_1 \alpha_2$ and $\sigma_2 = \tilde{\sigma}_2 \alpha$